SayedMorteza Malaekeh

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Research Interests

Geospatial and multimodal machine learning, remote sensing, and causal inference for Earth and environmental systems. Large-scale spatial, imagery, and administrative data for climate resilience, ecosystem valuation, and sustainable energy.

Education

University of Texas at Austin (UT Austin)
PhD Candidate (ABD) in Civil Engineering (Sustainable Systems) — *Jan. 2023 – Present*
Dissertation: *"Large-Scale Causal Inference for Sustainable Systems and Economic Policy"*
PhD Committee: Sergio Castellanos (UT Austin); Hannah Druckenmiller (Caltech); Connor Jerzak (UT Austin); Javad Mohammadi (UT Austin); Atila Novoselac (UT Austin); Emre Kıcıman (Microsoft)

University of Texas at Austin (UT Austin)
M.Sc. in Economics (full Ph.D. sequence, 12 courses) — *May 2026*

California Institute of Technology
Visiting PhD Researcher, Economics and Computer Science — *May–Aug. 2024; May–Dec. 2025*

Sharif University of Technology
M.Sc. in Water Resources Engineering (Thesis A+; Ranked First) — *Sep. 2019 – Feb. 2022*
Thesis: *"Spatio-temporal Variation of Climate Change and Its Impacts on Iranian Agriculture"*

Saint Petersburg State University
Exchange Master's Student, Applied Mathematics and Control Processes — *Feb. 2021 – Jul. 2021*
GPA: 4.0/4.0

Sharif University of Technology
B.Sc. in Civil Engineering with Minor in Economics — *Sep. 2015 – Aug. 2019*

Selected Publications (Google Scholar: Profile)

Manuscripts Under Review

[1] **Malaekeh, S.**, Castellanos, S., Barbose, G., & O'Shaughnessy, E. **Residential Solar Adoption Leaves Financially Constrained Households at Greater Risk.**
Under review, *Nature Energy*

Working Papers

[1] **Malaekeh, S.**, Druckenmiller, H., Jerzak, C., & Gkioxari, G. **A Multimodal Causal Framework for Estimating Location-Specific Ecosystem Values at Scale.** [Working Paper]
Workshop version: NeurIPS 2024, Tackling Climate Change with ML (peer-reviewed).

Peer-Reviewed Journal Articles

[1] **Malaekeh, S.**, French, J., Potts, J., & Castellanos, S. (2026). Multidimensional risk and resilience score: Informing utility investments in vulnerable communities. *Environmental Research Communications*, 8(6), 065009. [Article]

[2] **Malaekeh, S.**, & Castellanos, S. (2025). Residential solar photovoltaic seeding effects and their unequal impact on diffusion across racial and ethnic groups in the U.S. *Energy Policy*, 206, 114782. [Article]

[3] **Malaekeh, S.**, Shiva, L., & Safaie, A. (2024). Investigating the economic impact of climate change on agriculture in Iran: Spatial spillovers matter. *Agricultural Economics*, 55(3), 433–453. [Article] [Replication]

[4] Danesh-Yazdi, M., & **Malaekeh, S.** (2023). Investigating water resources and agricultural economy in the Mahabad study area by hydro-economic modeling. *Iran-Water Resources Research*, 19(2), 87–105. [Article]

[5] **Malaekeh, S.**, Safaie, A., Shiva, L., & Tabari, H. (2022). Spatio-temporal variation of hydro-climatic variables and extreme indices over Iran based on reanalysis data. *Stochastic Environmental Research and Risk Assessment*, 36, 3725–3752. [Article]

Academic and Research Appointments

Graduate Research Assistant | University of Texas at Austin, RESET Lab Jan. 2023 – Present

- Conduct research at the intersection of sustainable energy systems, environmental economics, and causal inference using large-scale administrative, household, and geospatial data.
- Study clean-energy technology adoption, household welfare, infrastructure resilience, and distributional impacts using econometric and causal machine-learning methods.

Visiting PhD Researcher | California Institute of Technology May–Aug. 2024; May–Dec. 2025

- Developed methodological advances in causal machine learning for non-tabular and spatial data, combining deep-learning representations with heterogeneous-treatment-effect estimation.
- Developed methods for spatial interference and multimodal causal inference and applied them to wetland loss, ecosystem valuation, and flood-mitigation benefits using geospatial imagery and tabular data.

Graduate Research Assistant | Lawrence Berkeley National Laboratory June 2024 – June 2025

- Led the design of a national study of residential solar adoption and household financial outcomes, including acquisition of a $180K Experian credit dataset covering approximately 10M households.
- Built scalable research pipelines for 13+ TB of longitudinal household credit data and applied staggered difference-in-differences, doubly robust estimation, and matching to estimate heterogeneous financial impacts.

Additional Research and Industry Experience

PhD Data Science Intern | Roblox Corporation May – Aug. 2026

- Redesigned ranking objectives for the Roblox Moments short-form video recommender system to optimize long-term retention rather than short-term engagement.
- Applied double machine learning, causal forests, deep learning, Bayesian optimization, and large-scale experimentation to petabyte-scale behavioral data, improving the long-term-retention ranking objective by 30%+.

Quantitative Research Fellow | Quanta Ventures Fund Jan. – Mar. 2026

- Developed and backtested machine-learning-based quantitative trading strategies using high-frequency financial data, including mean-reversion, momentum, and regime-based signals.

Teaching Experience

Instructor / Workshop Leader

- GIS & Remote Sensing Applications for Economists — Tehran Institute for Advanced Studies, Nov. 2022
- GIS & Remote Sensing — Sharif University of Technology, Dec. 2021
- Spatial Econometrics — Saint Petersburg State University, Spring 2021
- Introduction to Programming with MATLAB — Sharif University of Technology, 2018–2021

Teaching Assistant

- Building Environmental Systems — University of Texas at Austin, 2026
- Introduction to Energy Systems — University of Texas at Austin, 2026
- Causal Inference & Machine Learning — University of Texas at Austin, 2025
- Engineering Probability & Statistics — Sharif University of Technology, Spring 2020
- Hydraulic Laboratory — Sharif University of Technology, Spring 2020
- Water & Wastewater Management — Sharif University of Technology, 2018–2021

Presentations

2026 American Solar Energy Society National Solar Conference (ASES, oral), Austin, TX [upcoming]

2025 U.S. Association for Energy Economics (USAEE, oral), Fort Worth, TX; Workshop in Environmental Economics and Data Science (TWEEDS, Causal AI panel), Portland, OR

2024 NeurIPS Workshop on Tackling Climate Change with ML [link]; American Geophysical Union (AGU, oral), Washington, DC; Sloan/Berkeley Summer School in Environmental & Energy Economics (Egg Timer), UC Berkeley; Macro Energy Systems (MES), Princeton

2023 American Geophysical Union (AGU), San Francisco; Seminar in Water Economics OnLine (SWELL)

2022 European Geosciences Union (EGU, oral), Vienna

2021 European Geosciences Union (EGU), online

Fellowships, & Awards

2026 **Academy Distinguished Alumni Graduate Fellowship**, UT Austin
2025 **Robert Abbasi & Shahnaz Hemmati Graduate Fellowship**, UT Austin
2024 **Grad Camp Fellowship**, UC Berkeley & **Travel Grant**, Princeton University
2020 **Exchange Program Scholarship**, Sharif Univ. of Technology
2019 **Ranked 9th/10,000 (top 0.1%)** — Iranian National Economics Olympiad

Academic and Professional Service

Program Committee / Reviewer: NeurIPS Workshop on Generative AI for Health
Journal Reviewer: *Agricultural Economics*; *Theoretical and Applied Climatology*; *Environmental Monitoring and Assessment*
Board of Directors: Persian Student Society, University of Texas at Austin
Peer Mentor: University of Texas at Austin
Volunteer Teacher: Yarigaran Education Charity Group

Methods and Computing

Geospatial & Remote Sensing: GeoPandas, Rasterio, Google Earth Engine, GIS, remote sensing

Causal Inference & Econometrics: Difference-in-Differences, Event Studies, Double/Debiased Machine Learning, R-Learner, Causal Forests, Synthetic Control, Matching/IPW, Regression Discontinuity, Spatial Econometrics, A/B Testing

Machine Learning: Deep Learning, Vision Transformers, CNNs, Multimodal Learning, Representation Learning, XGBoost, Bayesian Optimization

Programming & Data: Python (PyTorch, Scikit-learn), R, SQL, STATA, MATLAB, Spark/PySpark, Hive, Trino, Databricks, HPC

References

Sergio Castellanos
Assistant Professor
Fariborz Maseeh Department of Civil, Architectural & Environmental Engineering
University of Texas at Austin
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Hannah Druckenmiller
Assistant Professor of Economics; William H. Hurt Scholar
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